

Spatial Analysis & ML Modeling of Crash Severity in California

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Contents

0.1	1. Introduction	1
0.2	2. Methodology Setup	1
0.3	3. Data Loading	1
0.4	4. Interactive Map	2
0.5	5. Machine Learning (Random Forest)	3
0.6	6. Spatial Heterogeneity (GWR)	4
0.7	7. Conclusion	6

0.1 1. Introduction

This report investigates the determinants of traffic accident severity in California using a hybrid approach of **Spatial Statistics** and **Machine Learning**.

0.2 2. Methodology Setup

Loading necessary libraries.

```
library(tidyverse)
library(sf)
library(sp)
library(GWmodel)
library(leaflet)
library(ranger)
library(viridis)
library(ggplot2)
```

0.3 3. Data Loading

Loading and cleaning the California dataset.

```

# Load Data (Check if file exists)
if(file.exists("US_Accidents_March23.csv")){
  raw_data <- read_csv("US_Accidents_March23.csv", show_col_types = FALSE)
} else {
  stop("File not found! Please check US_Accidents_March23.csv")
}

# Rename Columns if needed (Handling different naming conventions)
if("Temperature(F)" %in% names(raw_data)) {
  colnames(raw_data)[which(names(raw_data) == "Temperature(F)")] <- "Temperature"
}
if("Humidity(%)" %in% names(raw_data)) {
  colnames(raw_data)[which(names(raw_data) == "Humidity(%)")] <- "Humidity"
}
if("Visibility(mi)" %in% names(raw_data)) {
  colnames(raw_data)[which(names(raw_data) == "Visibility(mi)")] <- "Visibility"
}

# Filter & Feature Engineering
df_ca <- raw_data %>%
  filter(State == "CA") %>%
  filter(!is.na(Start_Lat) & !is.na(Start_Lng)) %>%
  filter(!is.na(Severity) & !is.na(Temperature)) %>%
  select(ID, Severity, Start_Lat, Start_Lng, Start_Time,
         Temperature, Visibility, Humidity,
         Weather_Condition, Junction, Crossing) %>%
  mutate(
    Start_Time = as.POSIXct(Start_Time, format="%Y-%m-%d %H:%M:%S"),
    Hour = as.numeric(format(Start_Time, "%H")),
    Is_Night = ifelse(Hour >= 19 | Hour <= 6, 1, 0),
    Is_Severe = ifelse(Severity >= 3, 1, 0),
    Is_Severe_Factor = as.factor(Is_Severe)
  ) %>%
  na.omit()

# Clear memory
rm(raw_data)
gc()

```

```

##           used (Mb) gc trigger (Mb) max used (Mb)
## Ncells  4851499 259.1   21058540 1124.7 15792130  843.4
## Vcells 39790743 303.6   541603064 4132.2 632108241 4822.7

```

```
print(paste("Total Cleaned Records:", nrow(df_ca)))
```

```
## [1] "Total Cleaned Records: 1684869"
```

0.4 4. Interactive Map

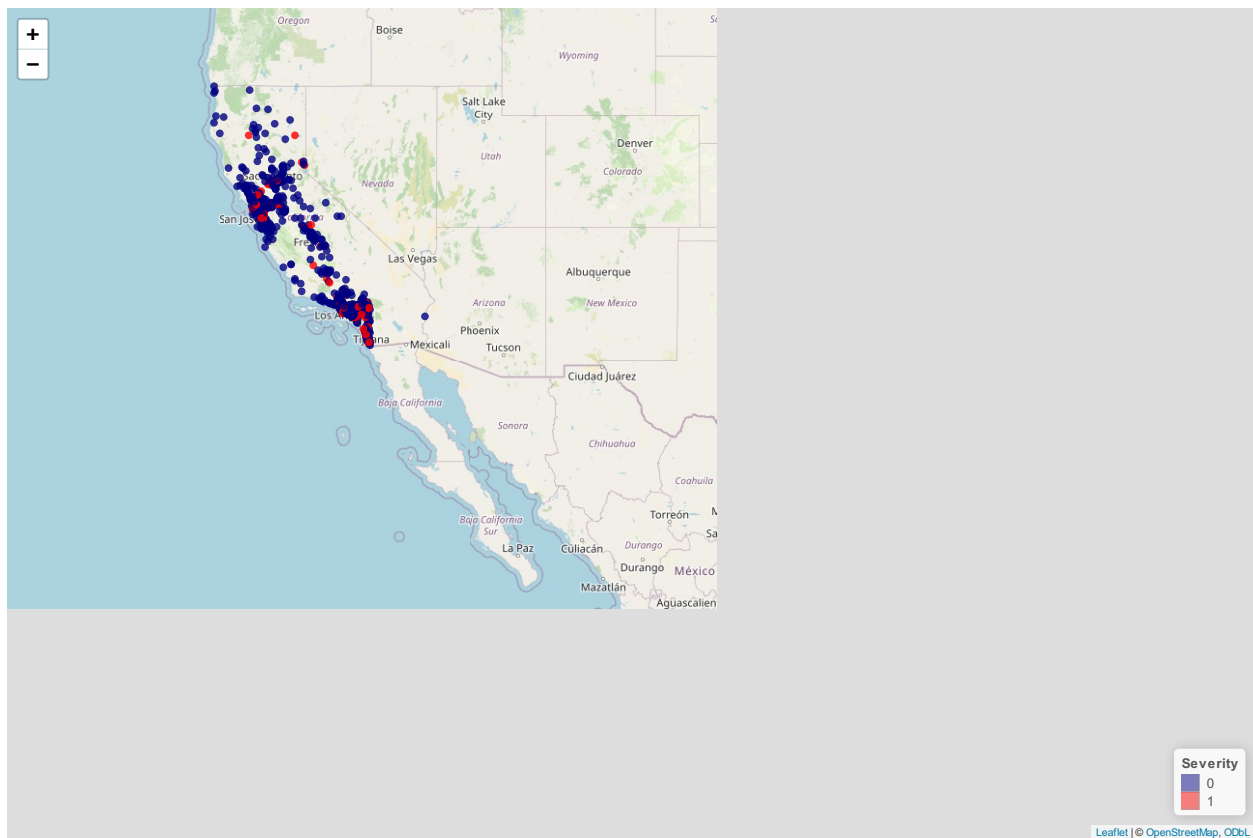
Visualizing crash hotspots (Sample of 1000 points).

```

set.seed(123)
map_sample <- df_ca %>% sample_n(1000)
pal <- colorFactor(c("navy", "red"), domain = c(0, 1))

leaflet(map_sample) %>%
  addTiles() %>%
  addCircleMarkers(
    ~Start_Lng, ~Start_Lat,
    color = ~pal(Is_Severe),
    radius = 4, stroke = FALSE, fillOpacity = 0.8,
    popup = ~paste("Severity:", Severity, "<br>Temp:", Temperature)
  ) %>%
  addLegend("bottomright", pal = pal, values = ~Is_Severe,
            title = "Severity", labels = c("Minor", "Severe"))

```



0.5 5. Machine Learning (Random Forest)

Identifying key factors affecting severity.

```

set.seed(42)
ml_data <- df_ca %>% sample_n(min(20000, nrow(df_ca)))

rf_model <- ranger(Is_Severe_Factor ~ Temperature + Visibility + Humidity +
  Hour + Is_Night + Junction + Crossing,
  data = ml_data,

```

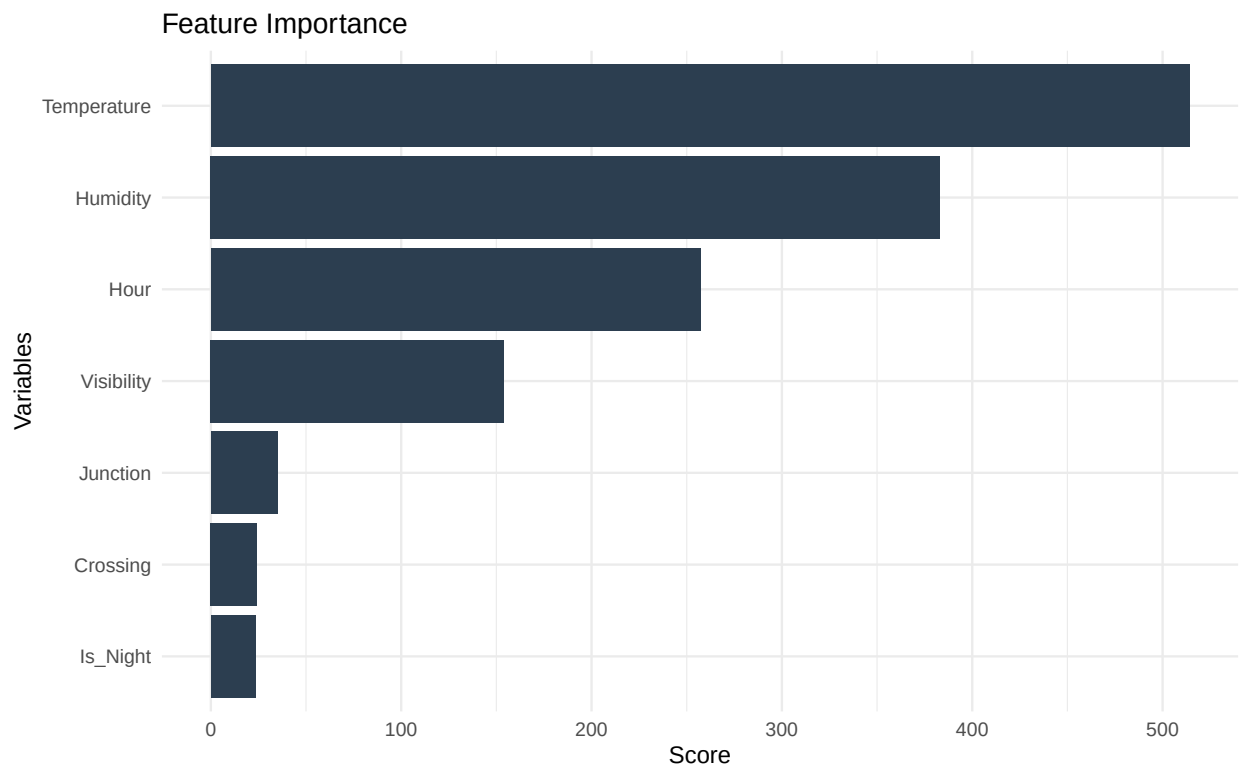
```

importance = 'impurity',
num.trees = 100)

imp_df <- data.frame(Variable = names(rf_model$variable.importance),
                     Importance = rf_model$variable.importance)

ggplot(imp_df, aes(x = reorder(Variable, Importance), y = Importance)) +
  geom_bar(stat = "identity", fill = "#2c3e50") +
  coord_flip() +
  theme_minimal() +
  labs(title = "Feature Importance", x = "Variables", y = "Score")

```



0.6 6. Spatial Heterogeneity (GWR)

Investigating how Temperature impact varies across space.

```

set.seed(99)
gwr_data <- df_ca %>% sample_n(3000)
sp_data <- gwr_data
coordinates(sp_data) <- ~Start_Lng+Start_Lat

# GWR Model
bw <- bw.gwr(Is_Severe ~ Temperature, data = sp_data, approach = "AIC",
             kernel = "gaussian", adaptive = TRUE)

```

Take a cup of tea and have a break, it will take a few minutes.

```
##          -----A kind suggestion from GWmodel development group
## Adaptive bandwidth (number of nearest neighbours): 1861 AICc value: 2522.011
## Adaptive bandwidth (number of nearest neighbours): 1158 AICc value: 2516.691
## Adaptive bandwidth (number of nearest neighbours): 722 AICc value: 2504.525
## Adaptive bandwidth (number of nearest neighbours): 454 AICc value: 2492.333
## Adaptive bandwidth (number of nearest neighbours): 287 AICc value: 2486.393
## Adaptive bandwidth (number of nearest neighbours): 185 AICc value: 2480.093
## Adaptive bandwidth (number of nearest neighbours): 120 AICc value: 2472.743
## Adaptive bandwidth (number of nearest neighbours): 82 AICc value: 2464.759
## Adaptive bandwidth (number of nearest neighbours): 56 AICc value: 2457.527
## Adaptive bandwidth (number of nearest neighbours): 42 AICc value: 2450.679
## Adaptive bandwidth (number of nearest neighbours): 31 AICc value: 2441.621
## Adaptive bandwidth (number of nearest neighbours): 27 AICc value: 2441.485
## Adaptive bandwidth (number of nearest neighbours): 21 AICc value: 2443.586
## Adaptive bandwidth (number of nearest neighbours): 27 AICc value: 2441.485
```

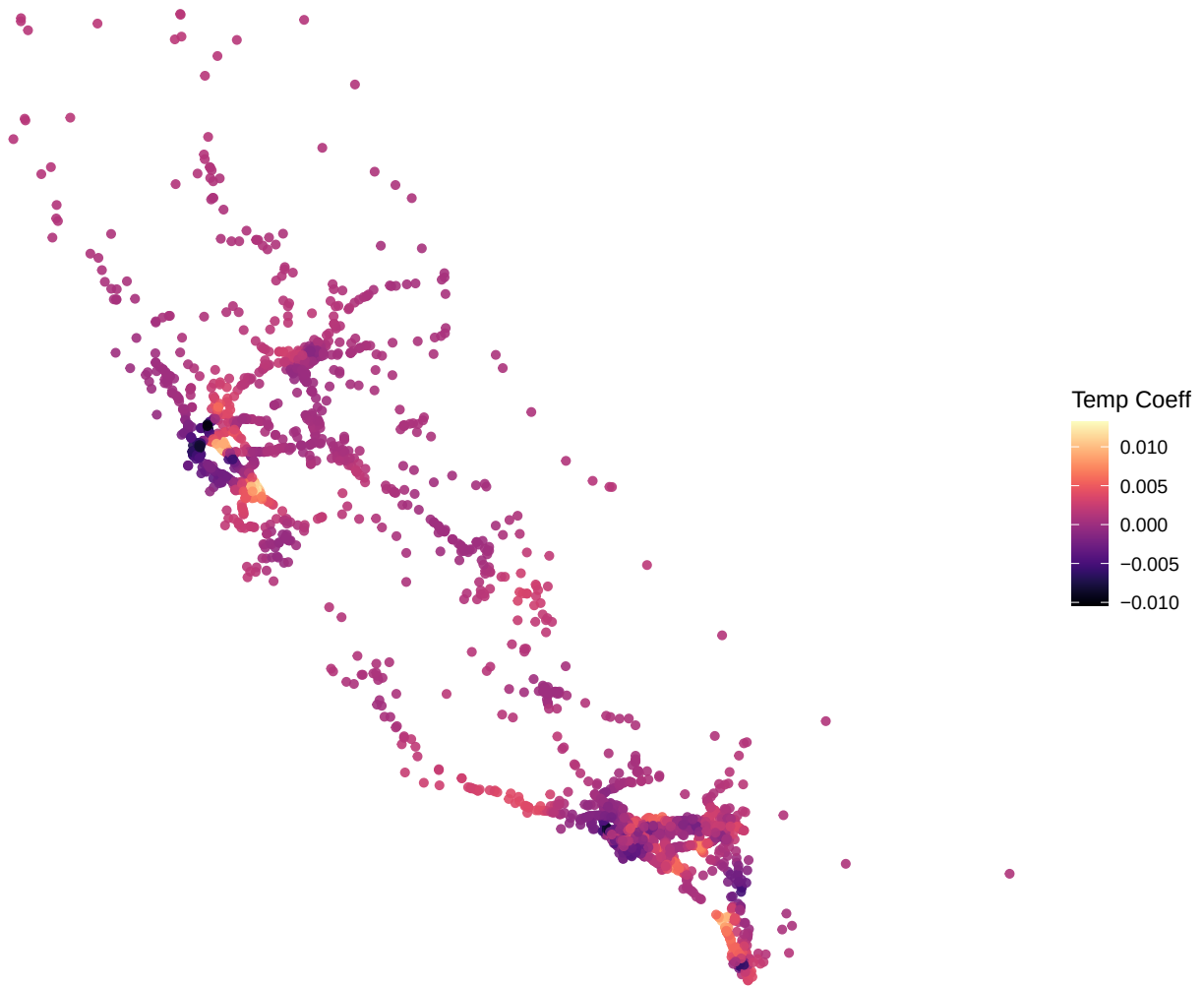
```
gwr_model <- gwr.basic(Is_Severe ~ Temperature, data = sp_data,
                      bw = bw, kernel = "gaussian", adaptive = TRUE)

# Plot Results
gwr_results <- as.data.frame(gwr_model$SDF)

ggplot() +
  geom_point(data = gwr_results,
            aes(x = Start_Lng, y = Start_Lat, color = Temperature),
            size = 1.5, alpha = 0.9) +
  scale_color_viridis(option = "magma", name = "Temp Coeff") +
  theme_void() +
  labs(title = "Spatial Heterogeneity: Temperature Impact",
       subtitle = "Yellow areas = Stronger impact") +
  coord_fixed()
```

Spatial Heterogeneity: Temperature Impact

Yellow areas = Stronger impact



0.7 7. Conclusion

The analysis confirms spatial heterogeneity in crash factors.